

Week #1

Meeting #1 (July 5th, 10am - 12:30pm)

Details:

- ☒ Find a dataset to use (choose one or more from recommendations on project description)

Recommended datasets for implementing and evaluating the inspection system:

- A dataset for benchmarking anomaly detection methods with a focus on industrial inspection - [MVTec AD](#)
- Images of defected production items - Kolektor Surface-Defect Dataset 2 ([KSDD2](#))
- A real-world industrial anomaly dataset - BeanTech Anomaly Detection Dataset (BTAD): [Paper](#), [Dataset](#).
- A dataset aimed at benchmarking visual defect detection methods in industrial metal parts manufacturing - Metal Parts Defect Detection Dataset ([MPDD](#))

To-Do list:

- ☒ Create teamwork agreement → Done
- ☒ Go through recommended training/documentation (if necessary) → Done (Ryan)
- ☒ Go through recommended training/documentation (if necessary) → Done (Isaac)
- ☒ Go through recommended training/documentation (if necessary) → Done (Dayana)
- ☒ Go through recommended training/documentation (if necessary) → Done (Kaitlyn)

Organize image dataset (Document information in one file, alongside images inside github) (Deadline: Wednesday @2pm)

- ☒ Decide on a Dataset → Done (All)
- ☒ Define/Document defect classes and labels → Done (All)
- ☒ Document part types, defect classes, and data balance. → Done (All)
- ☒ Document images and labels for use in the image classification. → Done (All)
- ☒ Create folder structure (PASS/FAIL or defect classes) → Done (Ryan)

Build a Single-Image Inspection Function (Deadline: Sunday @12pm)

- ☒ Image Standardization/preprocessing (size, color, lighting etc) → Reviewed (Dayana)
 - ☒ Image processing to extract defects (e.g. evidence mask) → Reviewed (Kaitlyn)
 - ☒ Small set of interpretable measurements to quantify defect evidence (number of suspicious regions, largest region area, fraction flagged) → Done (Isaac)
 - ☒ Completed single-frame evaluation model → Done (Isaac)
 - ☒ Single-frame evaluation model validation → Done (Ryan)
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Week #2

Meeting #3 (July 12th, 10am - 12:00pm)

To-Do list:

- ☒ ~~Integrate AI classifier into the single-frame evaluation model → Done (Issac)~~
 - ☒ ~~Fix Multiple tubes being classified as one object → Done (Ryan)~~
 - ☒ ~~Fix some tubes not being found → Done (Ryan)~~
 - ☒ ~~Improve classical inspection function performance → Done (Ryan)~~
 - ☒ ~~Implement basic batch processing → Done (Ryan)~~
 - ☒ ~~Adding a runtime to see how long it takes to train the data → Dropped~~
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Week #3

Meeting #5 (July 19th, 10am - 12:00pm)

Details:

- 1) N/A

To-Do list:

- ☒ ~~Remove image displaying from the single image inspection function, needs to return final label, confidence score, evidence overlay (picture), evidence metrics → Done (Isaac)~~
 - ☒ ~~Clean-up overall program (i.e. switch from live scripts to scripts, separate functionality into separate functions for better readability, try to improve system memory/runtime performance wherever possible) → Done (Isaac)~~
 - ☒ ~~Create a batch testing and training function → Done (Ryan)~~
 - ☒ ~~Perform robustness testing and generate their own individual results. (Brightness, Blur, Noise) → Done (Ryan)~~
 - ☒ ~~Create the template for the final report → Doing (Kaitlyn)~~
 - ☒ ~~Re-organize folders with a new naming scheme (folder renaming, images with unique titles, removal of the test/train folders) → Doing (Isaac)~~
 - ☒ ~~Clean-up program code (i.e. improve program runtime, comment code structure/format, etc.) → Doing (ALL)~~
 - ☒ ~~Organize GitHub repository (move program files to main branch, separate → Doing (Isaac)~~
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Week #5

Meeting #7 (July 26th, 10am - 12:00pm)

Details:

1) N/A

To-Do list:

- ☒ Finish the testing script → Done (Ryan)
- ☒ Readme for Github (instructions, short overview) → Done (Ryan)
- ☒ Delete other folder of un-used datasets → Done (Isaac)
- ☒ Agree on a model to upload → Done (ALL)
- ☒ Organize the repository, bring everything back to the main branch
(<https://github.com/mathworks/MATLAB-Simulink-Challenge-Project-Hub/blob/main/Glasroom%20Challenge%20Projects/README.md>) → Done (Isaac)
- ☒ Introduction → Done (Dayana)
- ☒ Background → Done (Dayana)
- ☒ Dataset → Doing (Kaitlyn)
- ☒ Methodology → Doing (Kaitlyn)
- ☒ Experimental Setup → Done (Isaac)
- ☒ Results → Doing (Isaac)
- ☒ Research and attend Tuesday's question session (2pm-3pm) to find out exactly what we need in our submission → Done (Kaitlyn)
 - How and when they will determine who has to present
 - Figure out how presentations will work
 - Figure out how much each person needs to know, or if everyone just needs to be an expert in one particular area
 - In detail, what do we need for our final submission (i.e. GitHub repository expectations, final report expectations, etc.)
- ☒ Potentially add a UI popout with the information rather than one in the output. → To-Do (Isaac)

To-Do list (Post Wednesday Meeting):

- ☒ Robustness testing (in Results section) → Done (Isaac)
- ☒ Conclusion → Done (Ryan)
- ☒ Abstract → Done (Isaac)
- ☒ Update Meeting Notes/To-Do list for github documentation → Doing (Ryan)
- ☒ Presentation? (No longer necessary)
- ☒ Dataset → Done (Kaitlyn)
- ☒ Methodology → Done (Kaitlyn)
- ☒ References → Done (Dayana)
- ☒ DOUBLE CHECK dataset citation in repository → Done (Ryan)

@INPROCEEDINGS{9631567,

```

author={Jezek, Stepan and Jonak, Martin and Burget, Radim and Dvorak, Pavel
and Skotak, Milos},
booktitle={2021 13th International Congress on Ultra Modern
Telecommunications and Control Systems and Workshops (ICUMT)},
title={Deep learning-based defect detection of metal parts: evaluating
current methods in complex conditions},
year={2021},
volume={},
number={},
pages={66-71},
doi={10.1109/ICUMT54235.2021.9631567}

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- ☒ ~~AFTERTHOUGHT/EXTRACURRICULAR: Add pop-up window to current program for “UI enhancement” → Dropped~~

Meeting #9 (August 2nd, 10am - 12:00pm)

Details:

- 1) N/A

To-Do list:

- ☒ ~~Potentially add an organized window for GF to match the single image inspection function → Dropped~~
- ☒ ~~Include popout instruction in readme → Done (Ryan)~~
- ☒ ~~Correct tenses/removes the words ‘us’ ‘our’ ‘we’ (i.e first person) from the paper. → Done (ALL)~~
- ☒ ~~Proofread Sections 1&2+ABSTRACT → Done (Ryan)~~
- ☒ ~~Proofread Sections 3&4 → Done (Isaac)~~
- ☒ ~~Proofread Sections 5&6 → Done (Kaitlyn)~~
- ☒ ~~Proofread Sections 7&8 → Done (Dayana)~~
- ☒ ~~Proofread references and Add ‘main’ function to Appendix (PDF format; all other scripts can be added as PDF links) → To-Do (Isaac)~~
- ☒ ~~Formatting → Done (Ryan)~~
- ☒ ~~Verify program setup procedure(s) in GitHub (i.e. remove local repository, clone current repository, then follow setup procedures and try to run the program, document improvements where necessary) → Done (Kaitlyn)~~
- ☒ ~~Bug fix for text color issue → Done (Isaac)~~

To-Do list (Post Wednesday Meeting):

- ☒ ~~Final Checks and submission → Done (ALL)~~